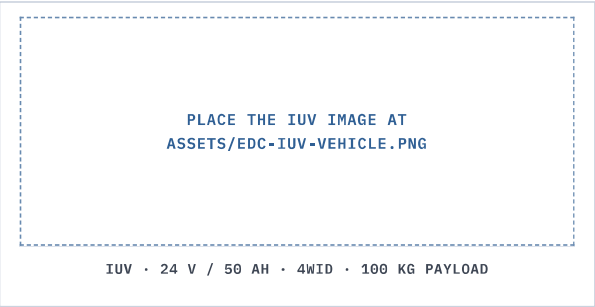




SENSORS & SAFETY INPUTS

2D LIDAR

Obstacle detection and mapping input.

IMU

Yaw-rate and acceleration measurements.

WHEEL ENCODERS ×4

Wheel-speed feedback and odometry.

E-STOP + BUMPER

Hardwired stop and collision chain.

SAFETY SENSORS

Front and rear obstacle-zone inputs.

MAIN CONTROLLER

INDUSTRIAL PC · UBUNTU 24.04 · ROS 2 JAZZY

SLAM / LOCALIZATION

MOTION PLANNER

DIAGNOSTICS

FLEET CLIENT

High-level autonomy, mission execution, navigation, map handling and system diagnostics.

FLEET MANAGEMENT / FACTORY IT

WI-FI / ETHERNET

MISSION DISPATCH

TRAFFIC

MAP + STATUS

Fleet supervision with API links to WMS / MES and the factory IoT environment.

REAL-TIME MOTION CONTROLLER

MCU / DEDICATED MOTION CONTROLLER

4WID KINEMATICS

WHEEL-SPEED CONTROL

STRAIGHT CORRECTION

ODOMETRY

Runs deterministic steering and wheel-control functions in a 100–500 Hz loop.

POWER & SAFETY DISTRIBUTION

24 V / 50 Ah battery → BMS → fuse / contactor → motor bus. DC/DC supplies 12 V and 5 V loads; E-stop interrupts traction power.

ACTUATION · DRIVE DRIVERS ×2

24 V MOTOR POWER STAGE

CAN

RS485

PWM

Two drive power stages command four wheel motors.

HUB MOTORS ×4

24 V · 25 A · 15 NM · 370 R/MIN

Four independently driven, fixed-orientation hub motors.

REFERENCE PARTITION

Industrial PC handles autonomy; an MCU handles deterministic motion; hardwired devices interrupt traction independently.

CONTENT REPRODUCED FROM THE EDC REFERENCE, NOT GVIEVO-01

